

market to drive those kind of resolutions with enough video RAM buffer to handle the graphics.

As you may have realised by now, the evolution of the TV and monitor landscape has a lot to do with the enabling eco-systems around them. We would still be on CRT TVs had TV broadcasters stuck to the analogue medium and Sony hadn't come out with the Blu-ray drive. So the main challenge for UltraHD or 4K technology is the lack of enabling eco-system, at the moment.

4K resolution equates to 3840x2160 pixels which is four times that of the current full HD resolution. The concern at this resolution is the availability of content. The main reason behind the failure of 3D TVs was the lack of native 3D content. While we are seeing a lot of TV companies such as Sony, Toshiba, LG and Samsung announcing their 84-inch 4K UHDTVs, only Sony so far has laid out a roadmap talking about 4K content delivery. We are not even getting into the content creation aspect of it, which comes with its own set of challenges. Sure there are digital cameras capable of shooting raw 4K content, but processing it and compressing it for consumption on a 4K TV is an altogether different beast.

On the content delivery side, Sony plans to launch a 4K media player and a 4K video distribution service – FMP-X1 – in the next few months. The \$699 FMP-X1 will connect with Sony's on-demand 4K movie download service to deliver 4K content. As of now Sony has announced

10 movie titles which are "mastered in 4K" (Blu-ray HD titles upscaled to 4K resolution) which will be available in July when its 4K TVs launch. Towards the end of the year, Sony also plans to launch its much-awaited PlayStation 4 which will be 4K compatible. The lack of a physical medium such as Blu-ray discs for carrying 4K content is another challenge because going by the current internet bandwidths, streaming 4K content over the internet or even 4K broadcasts is downright improbable. Advanced video compression technology such as High efficiency video coding (HEVC) will still require a bandwidth of 25-35 Mbps which is still a dream in most parts of the world for regular home broadband users.

The viewing experience on a 4K TV should throw up interesting results. As of now, the 4K TVs announced and showcased at various events abroad have been 65-inches plus, with 84-inches finding a common ground among most brands. At that screen size, native 4K content should look great and should give you a pixel-free viewing experience vis a vis 1080p content. But that is a prohibitively huge size of TV for the average Indian home. Sony recently announced its first 4K TV which starts from 55-inches which is still a manageable size. Assuming most households will have a sub-50-inch flatscreen TV in their house, it will be difficult to tell apart a 1080p and a 4K display if both are viewed from the same distance. If the content is full HD



Sony XBR-55X900A 4K Ultra HDTV

upscaled to 4K, as will be the case for the initial months, the image quality jump will not be that impressive. 4K computer monitors will still make a lot more sense as the viewing distance for a computer monitor is far less than that for a home TV.

The one area where 4K resolution will really work is in passive 3D displays. Current generation passive 3D displays tend to break up the vertical resolution into half such that the resolution for each eye is 1920x540 pixels. With a 3840x2160 pixel resolution, the breakup will be 3840x1080 for both the eyes thereby enabling you to view full HD resolution for both eyes.

So yes, be prepared for a lot of marketing blitz around the 4K technology in the coming months. Sony has already started taking orders for its 4K UHDTV abroad. Other brands should follow soon. Apart from hardcore videophiles and early adopters, we doubt the 4K segment will really see that many sales in India at least for the next couple of years.



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